

EXPERIMENT No :- 3

MariaDB [vijendra0862]> SELECT * FROM employee WHERE deptno = 30 order by sal;

empno	ename	job	mgr	hiredate	sal	comm	deptno
7900	JAMES	CLERK	7698	1981-12-03	950	NULL	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	300	30
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7499	ALLEN	SALESMAN	7698	1981-01-20	1600	300	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850	NULL	30

6 rows in set (0.001 sec)

1. ORDER by - Used to sort records in ascending descending order.

MariaDB [ashishsingh4431]> SELECT JOB, DEPTNO
-> FROM employee
-> WHERE ENAME LIKE 'A__N';

JOB	DEPTNO
SALESMAN	30

1 row in set (0.000 sec)

2. LIKE - Used for pattern matching (for many characters - for single characters)

3. IN - Used to match multiple values in a condition.

MariaDB [vijendra0862]> SELECT ename, job, deptno FROM employee WHERE ename Like "A__N" and length(ename) = 5;

ename	job	deptno
ALLEN	SALESMAN	30

1 row in set (0.001 sec)

4. OR - Combines multiple conditions

IS NOT NULL - checks for non-null values (eg- commission)

Arithmetic Expression - Used to calculate values like annual salary (sal * 12)

MariaDB [vijendra0862]> SELECT ename, job, deptno FROM employee WHERE ename Like "AS";

ename	job	deptno
JONES	MANAGER	20
ADAMS	CLERK	20
JAMES	CLERK	30

3 rows in set (0.001 sec)

- 3(i) List all employees and jobs in Department 30 in descending order by salary.

Maria DB [Ashish Singh] > SELECT * FROM employee
WHERE deptno = 30 order by sal
desc;

2. List job and Department number of employee whose name are five letters long begin with A and end with "N".

Maria DB [Ashish Singh] > SELECT ename, job
deptno FROM employee. where ename like "A...N"

✓ Maria DB [Ashish Singh] > SELECT ename, job, deptno
FROM employee where ename like "A%N"
and length (ename) = 5

3. Display the name of employees whose name start with alphabet S.

Maria DB [Ashish Singh] > SELECT job deptno
FROM employee where ename like "S%";

MariaDB [vijendra@862] > SELECT * FROM employee WHERE deptno IN (10,20,40) OR job IN ("CLERK", "SALESMAN", "ANALYST");

empno	ename	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	300	30
7566	JONES	MANAGER	7839	1981-04-02	2975	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7782	CLARK	MANAGER	7839	1981-06-09	2450	NULL	20
7788	SCOTT	ANALYST	7566	1982-12-09	3000	NULL	40
7839	KING	PRESIDENT	NULL	1981-11-17	5000	0	20
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100	NULL	20
7900	JAMES	CLERK	7699	1981-12-03	950	NULL	20
7902	FORD	ANALYST	7566	1981-12-03	3000	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300	NULL	10

13 rows in set (0.001 sec)

MariaDB [vijendra@862] > SELECT * FROM employee WHERE deptno IN (10,20,40) AND job IN ("CLERK", "SALESMAN", "ANALYST");

empno	ename	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800	NULL	20
7788	SCOTT	ANALYST	7566	1982-12-09	3000	NULL	40
7876	ADAMS	CLERK	7788	1983-01-12	1100	NULL	20
7902	FORD	ANALYST	7566	1981-12-03	3000	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300	NULL	10

5 rows in set (0.001 sec)

MariaDB [vijendra@862] > SELECT ename, empno, comm FROM employee WHERE comm IS NOT NULL AND comm > 0;

ename	empno	comm
ALLEN	7499	300
WARD	7521	300
MARTIN	7654	1400

3 rows in set (0.001 sec)

5. Display the name of employees with alphabet S.

MariaDB [Ashish@862] > SELECT ename, job, deptno FROM employee WHERE ename LIKE 'S%';

5. Display the names of employees working in department number 10 or 20 or 40 or employees working as clerks salesman or analyst.

MariaDB [Ashish@862] > SELECT * FROM employee WHERE deptno IN (10,20,40) OR job IN ("CLERK", "SALESMAN", "ANALYST");

MariaDB [Ashish@862] > SELECT * FROM employee WHERE deptno IN (10,20,40) AND job IN ("CLERK", "SALESMAN", "ANALYST");

6. Display employee number and name to employees who earn commission.

MariaDB [Ashish@862] > SELECT ename, empno, comm FROM employee WHERE comm IS NOT NULL AND comm > 0;


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MariaDB [vijendra0862]> SELECT empno, ename, sal + IFNULL(comm, 0) AS Totalsal FROM employee;
+-----+-----+-----+
| empno | ename  | Totalsal |
+-----+-----+-----+
| 7369   | SMITH  | 800      |
| 7499   | ALLEN  | 1900     |
| 7521   | WARD   | 1550     |
| 7566   | JONES  | 2975     |
| 7654   | MARTIN | 2650     |
| 7698   | BLAKE  | 2850     |
| 7782   | CLARK  | 2450     |
| 7788   | SCOTT  | 3000     |
| 7839   | KING   | 5000     |
| 7844   | TURNER | 1500     |
| 7876   | ADAMS  | 1100     |
| 7900   | JAMES  | 950      |
| 7902   | FORD   | 3000     |
| 7934   | MILLER | 1300     |
+-----+-----+-----+
14 rows in set (0.001 sec)

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Display employee number and total salary for each employee.

MariaDB [vijendra0862]> SELECT empno, ename, sal + IFNULL(comm, 0) AS Totalsal FROM employee;

MariaDB [vijendra0862]> SELECT empno, ename, sal + CASE WHEN comm IS NULL THEN 0 ELSE comm END AS Totalsal FROM employee;

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MariaDB [vijendra0862]> SELECT empno, ename, sal + CASE WHEN comm IS NULL THEN 0 ELSE comm END AS Totalsal FROM employee;
+-----+-----+-----+
| empno | ename  | Totalsal |
+-----+-----+-----+
| 7369   | SMITH  | 800      |
| 7499   | ALLEN  | 1900     |
| 7521   | WARD   | 1550     |
| 7566   | JONES  | 2975     |
| 7654   | MARTIN | 2650     |
| 7698   | BLAKE  | 2850     |
| 7782   | CLARK  | 2450     |
| 7788   | SCOTT  | 3000     |
| 7839   | KING   | 5000     |
| 7844   | TURNER | 1500     |
| 7876   | ADAMS  | 1100     |
| 7900   | JAMES  | 950      |
| 7902   | FORD   | 3000     |
| 7934   | MILLER | 1300     |
+-----+-----+-----+
14 rows in set (0.001 sec)

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Display employee number and annual salary for each employee.

MariaDB [vijendra0862]> SELECT empno, ename, (sal + IFNULL(comm, 0)) * 12 AS Annualsal FROM employee;

Display the names of all employee working as clerks and drawing a salary more than 3000.

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MariaDB [vijendra0862]> SELECT ename, (sal + IFNULL(comm, 0)) * 12 AS Annualsal FROM employee;
+-----+-----+
| ename  | Annualsal |
+-----+-----+
| SMITH  | 9600      |
| ALLEN  | 22800     |
| WARD   | 18600     |
| JONES  | 35700     |
| MARTIN | 31800     |
| BLAKE  | 34200     |
| CLARK  | 29400     |
| SCOTT  | 36000     |
| KING   | 60000     |
| TURNER | 18000     |
| ADAMS  | 13200     |
| JAMES  | 11400     |
| FORD   | 36000     |
| MILLER | 15600     |
+-----+-----+
14 rows in set (0.001 sec)

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MariaDB [vijendra0862]> SELECT ename, (sal + IFNULL(comm, 0)) * 12 FROM employee WHERE job = 'CLERK' and (sal + IFNULL(comm, 0)) * 12 > 3000;

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MariaDB [vijendra0862]> SELECT ename, (sal + IFNULL(comm, 0)) * 12 FROM employee WHERE job = 'CLERK' and (sal + IFNULL(comm, 0)) * 12 > 3000;
+-----+-----+
| ename  | Annualsal |
+-----+-----+
| SMITH  | 9600      |
| ADAMS  | 13200     |
| JAMES  | 11400     |
| MILLER | 15600     |
+-----+-----+
4 rows in set (0.001 sec)

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MariaDB [vijendra@0862] > SELECT ename, sal from employee where job in('CLERK', 'SALESMAN', 'ANALYST') AND sal >= 3000;

ename	sal
SCOTT	3000
FORD	3000

2 rows in set (0.001 sec)

10:- Display the name of employees who are working as clerk salesman or analyst and drawing a salary more than 3,000.

MariaDB [vijendra@0862] > SELECT ename, sal from employee where job in('CLERK', 'SALESMAN', 'ANALYST') AND sal >= 3000;